

ST PETERS, PE, Canada

WMO: 713100

Lat: 46.4503N

Lon: 62.5758W

Elev: 97

StdP: 14.64

Time Zone: -4.00 (NAA)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	4.2	-7.4	3.7	1.9	-3.5	4.6	5.5	29.6	18.8	27.3	19.9	11.4	240	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	81.7	69.8	79.0	68.0	76.2	66.7	71.9	79.0	70.2	76.1	68.5	73.9	11.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	107.9	75.9	67.9	103.0	74.5	66.4	97.6	72.4	35.7	79.5	34.2	76.1	32.9	74.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.1	22.1	19.8		DB	-5.0	86.3	3.2	1.7	-7.3	87.5	-9.2	88.5	-11.0	89.5	-13.3	90.7
				WB	-5.8	74.6	3.0	1.5	-8.0	75.7	-9.7	76.6	-11.4	77.4	-13.7	78.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	22.7	21.2	27.3	37.9	47.6	57.2	66.5	66.0	59.1	49.2	39.5	29.3
	DBStd	17.34	9.97	8.97	7.81	7.29	7.26	6.98	5.49	4.68	6.13	6.55	7.65	8.56
	HDD50	3947	845	806	706	374	136	13	0	0	4	95	327	641
	HDD65	7960	1310	1226	1170	814	540	251	48	44	195	491	765	1106
	CDD50	1670	0	0	1	10	62	229	513	496	277	70	12	0
	CDD65	207	0	0	0	0	1	16	96	75	18	1	0	0
	CDH74	1065	0	0	0	1	17	104	543	331	66	3	0	0
	CDH80	145	0	0	0	0	1	8	89	41	6	0	0	0
Wind	WSAvg	9.8	11.6	10.7	10.5	9.9	9.1	8.4	8.0	7.9	9.1	9.9	11.0	11.7
Precipitation	PrecAvg	47.6	4.1	3.4	3.5	3.3	3.2	3.2	3.4	3.9	4.2	5.0	4.6	5.3
	PrecMax	61.5	9.8	5.9	5.8	5.7	6.7	6.1	6.7	8.5	8.5	10.5	8.1	11.0
	PrecMin	38.7	1.4	0.7	1.3	1.5	1.9	1.4	0.8	1.2	1.1	1.8	2.1	2.6
	PrecStd	5.9	1.7	1.3	1.2	1.1	1.1	1.1	1.3	2.1	2.0	2.1	1.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	49.5	52.3	68.1	76.1	80.9	85.3	83.9	79.9	71.8	64.4	55.6
		MCWB	53.3	47.8	45.0	58.0	63.0	66.6	72.6	70.8	69.9	63.4	60.3	53.7
	2%	DB	45.8	42.5	45.8	60.9	69.9	77.2	82.5	80.7	75.5	66.5	59.8	50.0
		MCWB	44.8	40.0	42.1	54.3	58.1	65.2	70.8	69.4	67.3	60.8	57.3	48.5
	5%	DB	40.2	37.1	41.5	54.7	65.1	73.4	80.1	77.8	72.4	62.5	55.9	45.4
		MCWB	38.3	34.8	38.2	49.2	55.6	63.6	68.9	67.7	65.1	58.3	53.6	43.9
	10%	DB	35.6	33.0	37.8	49.8	60.5	69.5	77.0	74.9	69.3	59.2	51.5	40.9
		MCWB	34.1	31.4	34.7	45.1	53.3	61.6	67.4	66.5	63.7	55.3	49.0	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	47.6	47.7	60.8	65.2	71.1	74.2	73.4	71.6	66.3	61.3	54.0
		MCDB	55.4	49.5	50.1	65.7	73.8	77.5	82.9	80.1	77.5	69.6	64.0	55.1
	2%	WB	44.7	40.3	42.8	55.3	61.1	68.1	72.5	71.1	68.9	61.7	57.5	48.6
		MCDB	45.6	41.9	44.8	59.8	66.8	73.5	80.5	77.6	74.0	64.8	59.6	49.8
	5%	WB	38.6	35.0	38.5	50.6	57.2	65.2	70.8	69.5	66.7	58.7	53.9	44.0
		MCDB	40.1	36.8	41.2	53.7	62.2	71.0	77.4	75.1	70.7	62.2	55.6	45.2
	10%	WB	34.3	31.6	34.8	45.2	53.7	62.1	69.0	67.9	64.5	55.7	49.1	39.5
		MCDB	35.4	32.9	37.5	49.3	59.6	68.5	74.6	73.3	68.2	58.8	51.2	40.6
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.0	12.4	13.5	15.5	14.8	14.9	14.2	14.1	11.9	11.3	10.9
		MCDBR	19.5	17.8	18.2	23.0	24.8	21.4	19.6	18.7	18.4	16.9	17.3	17.1
		MCWBR	19.6	16.1	15.3	17.0	15.3	12.1	10.3	9.9	11.6	13.1	16.7	17.6
	5% WB	MCDBR	19.8	17.9	17.4	21.7	21.4	18.2	16.4	15.8	15.6	15.5	17.2	17.0
		MCWBR	20.1	17.0	15.9	18.0	15.7	11.9	9.7	9.6	11.3	13.4	17.3	17.6
Clear-Sky Solar Irradiance	taub	0.299	0.289	0.319	0.349	0.368	0.383	0.409	0.397	0.358	0.338	0.319	0.295	
	taud	2.360	2.289	2.296	2.403	2.429	2.403	2.370	2.404	2.502	2.522	2.490	2.458	
	Ebn at Noon	252	280	287	288	285	280	271	270	272	262	245	242	
	Edh at Noon	23	30	35	35	35	36	37	35	29	24	21	19	
All-Sky Solar Radiation	RadAvg	325	545	984	1333	1629	1772	1800	1581	1204	740	378	252	
	RadStd	20	79	98	121	143	131	119	97	106	75	47	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page